Context-aware Testing

28-04-2026

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Motivation

- Long term preferences/states: „vegetarian“, „favourite eating place“, „home Address“
- Overreliance habits:
  - „I usually prefer to make stops during trips.“ - > „..“
  - “Navigate to work.“ - > „Added a stop at …“
- Changed states:
  - „Save A as my home address.“ à „“
  - „Change my home address to B.“ à“ “
  - „Drive me home.“ à „Driving you to A.“

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Tasks

- Understand which contextual information relevant for Navigational Use Case and in General.
- Extension of STELLAR-ConvNavi to memorize contextual information
- Automated Benchmarking Contextual Understanding

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Approach

Inject context:
- Venue:restaurant
- Price:cheap

Simulate conversations and test context awareness:
- Conv1: I like only vegetarian food.
- Conv2: Find me a restaurant. -> Vegetarian restaurant
- Conv20: Find me a restaurant. -> Vegetarian restaurant

Simulate Timestamps
Minimize number of conversations
Maximize failures

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Related Work

- GitHub - xiaowu0162/LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory (ICLR 2025) · GitHub

- GitHub - MINE-USTC/VehicleMemBench: Official codebase for VehicleMemBench · GitHub

- Evaluating Memory in LLM Agents via Incremental Multi-Turn Interactions Yuanzhe Hu*†*†, Yu Wang*†, Julian McAuley

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Challenges

- Cost/Time for Conversation Simulation

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Challenges

- **Preference not retrieved:** dietary issues, vegan vs. vegetarian
- **Contradictory memory**: conflicting constraints unresolved (fastest vs. avoid highways)
- **Irrelevant recall**: unrelated past info influences routing (restaurant preference affects gas stop)
- **Over-personalization**: habits over-applied (automatic coffee stop on all trips)
- **Recall failure**: saved places not retrieved (e.g., “gym” not found)
- **Privacy leakage**: sensitive destinations exposed (e.g., clinic visits shown)
- **Temporal mismatch**: outdated info used after changes (old office location)
- **User confusion**: mixing preferences across drivers/in shared vehicles

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Vision

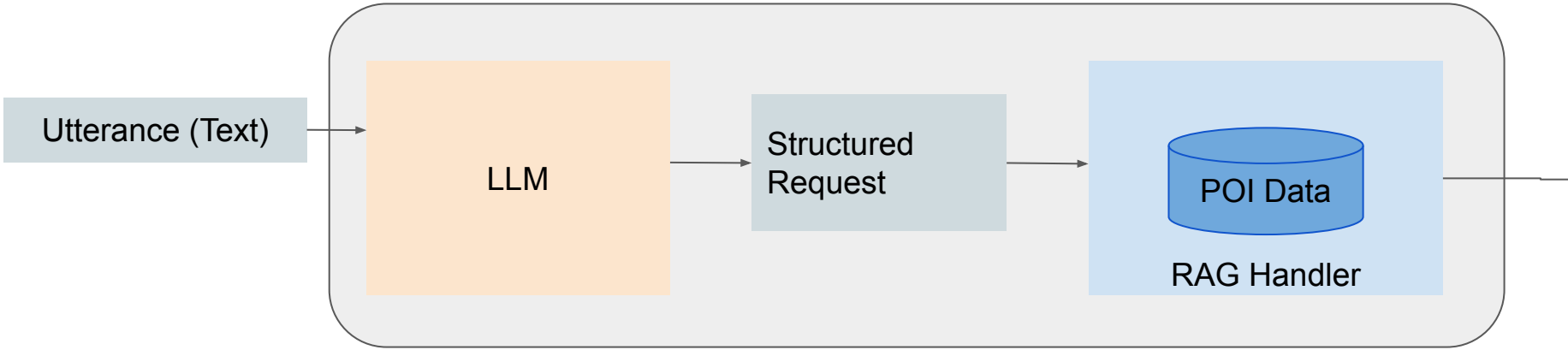
STELLAR: single-turn
LUNAR: multi-turn
LUNAR+: multi-conversation

We need:
- generic framework: change SUT, change case study
- dynamic framework (dynamic input generation)
- no/little initial data
- applicable to test in-car assistant
- efficient/effective

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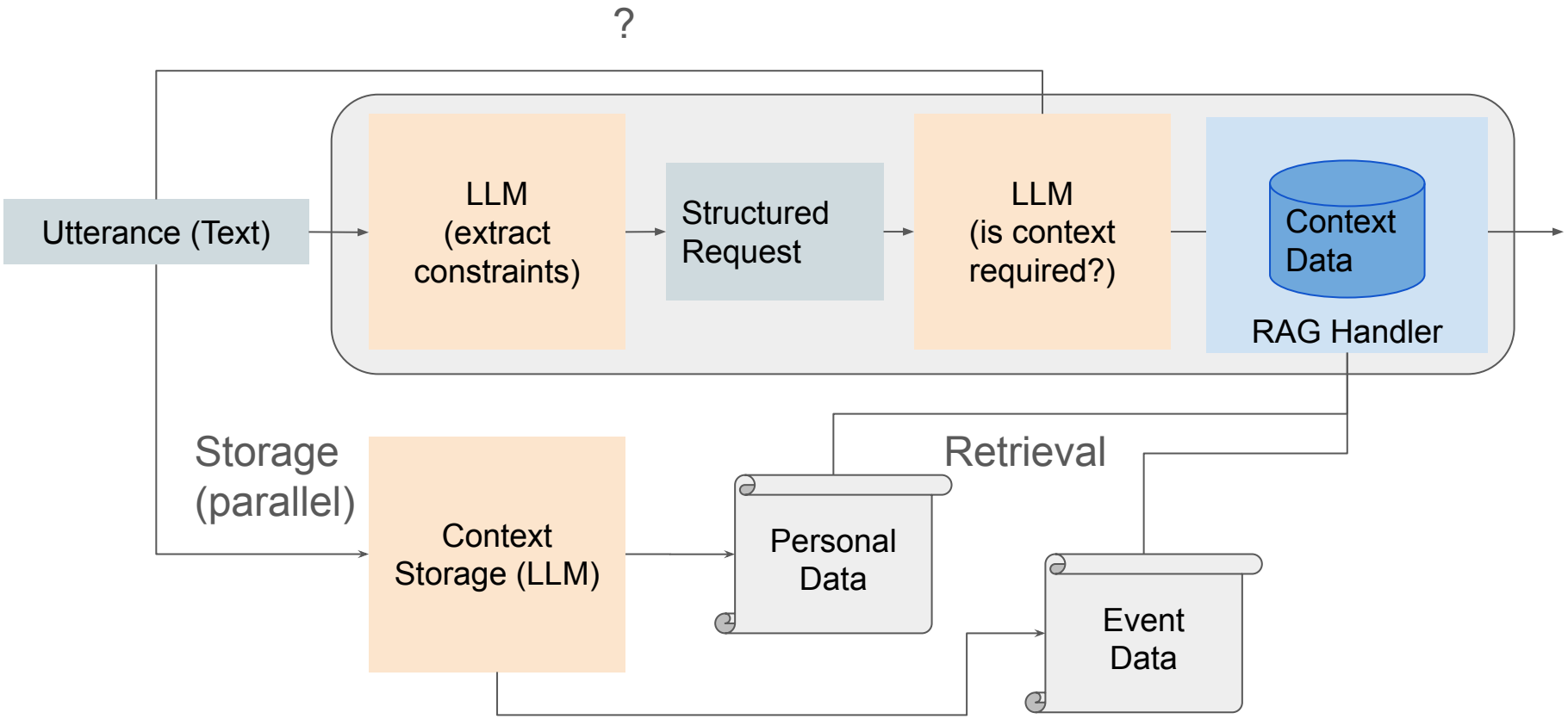
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Context Memory



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Context Memory



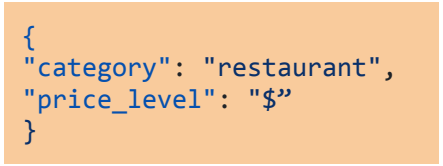
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I need a cheap restaurant for lanch.
Sorry I did not understand you.

Fail





Need to eat something for lunch.
I got it. Do you want directions to Pizzeria Franc?





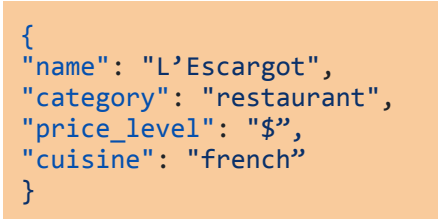
No, I need something French.
I have found the bar La Musique.

Fail





I need a restaurant.
There is L’Escargot, a nice French restaurant.



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Personal Data

- Name
- Age
- Religion
- Nationality
- Birth Place
- Home Place
- Friends
- Best Friend

…

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Preferences

- Food
- Bar
- Restaurant
-

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Event Data

- Timestamp
- Category
- Description

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Data Dimensions

Long-Term Volatile
Long-Term Non-Volatile
Short-Term

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Context-Aware Response

User Query
↓
Constraint Extraction (LLM)
↓
Memory Retrieval Decision
↓
Retrieve Relevant Memory
↓
Constraint Enrichment / Update
↓
POI Retrieval
↓
LLM Response Generation

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Context-Aware Storage

User Query

↓

Check if Memory Update required

↓

Update Memory

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